

Inertial Response from Variational Openness: A Generalized Mach Principle

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ABSTRACT

Mach's principle suggests that inertial response should not be regarded as an entirely intrinsic property of a local body, but as a consequence of its relation to the larger physical system in which it is embedded. We formulate this possibility through variational openness: the local variational balance retains a contribution from external degrees of freedom across an active interface. After causal reduction to a retained collective coordinate, this contribution appears as a temporal response whose low-frequency curvature contains a local acceleration sector. When the corresponding second temporal moment exists, the induced inertial coefficient equals one half of that moment. A passive harmonic realization yields a nonnegative induced contribution, while an independent Schur-complement condition controls stability. Within a fixed reduced description, the observed inertia is decomposed into residual local and openness-induced sectors, while the stronger claim that the residual sector vanishes is treated as an additional falsifiable hypothesis.

1. Introduction

In Newtonian mechanics, inertial mass is introduced operationally as the coefficient relating force to acceleration. This specifies how inertia is measured, but not why a local body should possess such a coefficient. Mach questioned the attribution of inertial properties to a body considered in complete isolation, arguing instead that they should reflect its relation to the larger physical system [1]. This relational intuition influenced Einstein's early discussion of gravitation and inertia [2], although it did not lead to a unique mathematical formulation of Mach's principle.

Later programmes implemented different aspects of this intuition. Sciamma proposed gravitational induction as a mechanism for inertia [3]; Brans and Dicke introduced a scalar-tensor theory partly motivated by Machian considerations [4]; Wheeler and Feynman showed how global absorber conditions can determine local radiative response [5]; and relational mechanics replaced absolute inertial structure by relations in configuration space [6]. Their diversity, emphasized in later classifications of Machian principles [7], shows that the Machian question is broader than any single interaction law.

The present Letter isolates a structural question common to these approaches: can a local inertial coefficient arise from the causal response of degrees of freedom excluded from a local variational description? Let a retained coordinate $q(t)$ receive, after reduction, a linear time-translation-invariant coordinate-memory response,

$$\begin{aligned}\mathcal{F}_K[q](t) &= \int_0^\infty K(s) q(t-s) ds, \\ m_{\text{open}} &= \frac{1}{2} \int_0^\infty s^2 K(s) ds.\end{aligned}\tag{1}$$

The second relation applies when the temporal moment exists and the retained motion is slow relative to the memory

timescale. The local derivative hierarchy of $q(t-s)$ then separates static, first-order, and acceleration sectors, so that the second temporal order contributes to the coefficient multiplying $\ddot{q}$. More generally, the same sector is identified by the low-frequency curvature of the retarded response. Memory generated by dynamical reduction provides the natural background for the present construction. Projection-operator methods derive memory kernels for retained variables [8, 9], while optimal prediction offers a related framework for reduced dynamics [10]. Influence-functional approaches likewise encode the effect of eliminated environmental degrees of freedom through temporally nonlocal effective dynamics [11, 12].

Building on these foundations, the present Letter establishes a direct correspondence between the quadratic low-frequency sector of a *coordinate*-memory response and an induced inertial coefficient. This coefficient is interpreted as the reduced dynamical imprint of variational openness and is embedded in a graded Machian framework with an explicit decoupling test. The construction also distinguishes two complementary routes to effective inertia: a kernel acting on q contributes through its second temporal moment, whereas a friction-memory kernel acting on $\dot{q}$ contributes through its first temporal moment [13].

The argument proceeds from a generalized open variational balance to a low-frequency inertial sector, a passive spectral realization, and an operational decoupling test. A covariant derivation of the retarded response from explicit exterior fields is left to a subsequent stage of the programme.

2. Generalized open variational balance

Let $\mathcal{S}$ denote the subsystem retained in the local description and $\mathcal{U}$ the larger physical system in which it is embedded. A retained trajectory q is *variationally open* when its realized dynamics is not selected by variation of the local action alone.

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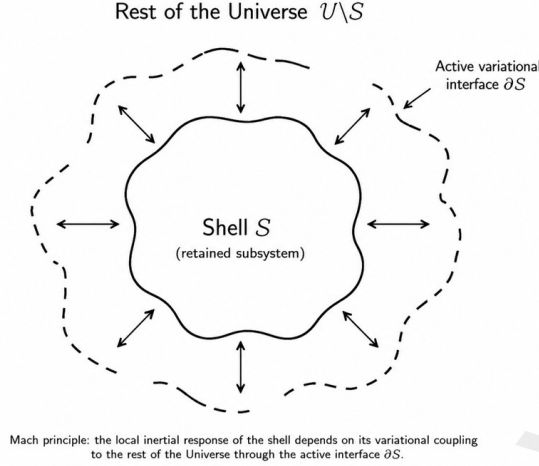


Figure 1: Open-shell representation of the generalized Mach principle. The retained subsystem S is variationally coupled to the complementary degrees of freedom $\mathcal{U} \setminus S$ across the active interface ∂S . The arrows denote causal variational coupling rather than material exchange. Reduction of this exterior response may generate memory and an induced inertial sector.

We write the open variational balance as

$$\delta S_S[q; \delta q] + B_{\mathcal{U} \rightarrow S}[q; \delta q] = 0. \quad (2)$$

The exterior contribution $B_{\mathcal{U} \rightarrow S}$ is a variational one-form: it is linear in the admissible variation δq , may be nonlocal in the trajectory, and represents the contribution that would be discarded under variational closure [14]. It is therefore not introduced as an arbitrary phenomenological force. It need not describe mechanical contact, be instantaneous, or be globally integrable as the variation of a scalar action.

If the exterior one-form is exact on trajectory space, one may write $B_{\mathcal{U} \rightarrow S} = \delta S_{\partial S}$, and Eq. (2) reduces to the open Hamilton form $\delta(S_S + S_{\partial S}) = 0$. The generalized balance is broader because reduced exterior responses may contain memory, dissipation, or gain and therefore need not derive from a single local action.

Here the term *boundary* refers to a variational partition rather than necessarily to a material surface. The active interface ∂S separates retained from complementary degrees of freedom and may be spatial, spacetime, configurational, or modal. It should not be confused with the exterior functional $B_{\mathcal{U} \rightarrow S}$. The corresponding shell–interface–exterior relation is summarized in Fig. 1.

After projection onto the retained coordinate, separate the exterior one-form into resolved time-local and unresolved retarded sectors. In the linearized reduction used below,

$$B_{\mathcal{U} \rightarrow S}[q; \delta q] = - \int \left[\Gamma_0 \dot{q}(t) - F_{\text{drive}}(t) + F_{\partial}^{\text{mem}}[q](t) \right] \delta q(t) dt.$$

The sign convention is chosen so that a positive resolved drive appears on the right-hand side of the dynamical equation. The coefficient Γ_0 represents the resolved time-local response, whereas $F_{\partial}^{\text{mem}}$ contains the unresolved history dependence.

Causality restricts this functional to the accessible past; representing it as a closed functional of the history of q is an additional coarse-grained assumption.

Linearization about a reference trajectory q_* defines the retarded response kernel through

$$K(t - t') = \frac{\delta F_{\partial}^{\text{mem}}[q](t)}{\delta q(t')} \Big|_{q=q_*},$$

$$K(t - t') = 0 \quad \text{for } t' > t.$$

Under time-translation invariance,

$$F_{\partial}^{\text{mem}}[q](t) = F_{\partial}^{\text{mem}}[q_*](t) + \int_{-\infty}^t K(t - t') [q(t') - q_*(t')] dt' + O((q - q_*)^2).$$

Taking q to denote the deviation from q_* and absorbing the reference response and homogeneous exterior transients into the resolved drive gives

$$m_0 \ddot{q}(t) + \Gamma_0 \dot{q}(t) + \kappa_0 q(t) + \int_0^\infty K(s) q(t-s) ds = F_{\text{drive}}(t). \quad (3)$$

Here m_0 and κ_0 are residual conservative coefficients of the local action, Γ_0 is the resolved time-local exterior response, and F_{drive} collects explicitly resolved forcing. Causality requires $K(s) = 0$ for $s < 0$; existence of temporal moments and validity of a local derivative expansion additionally require regularity and separation between the memory and retained-motion timescales.

3. Memory-induced inertial response

The primary definition used here is spectral. With the retarded frequency response

$$\tilde{K}(\omega) = \int_0^\infty K(s) e^{i\omega s} ds,$$

the induced inertial sector is defined whenever $\text{Re } \tilde{K}(\omega)$ is twice differentiable at zero frequency:

$$m_{\text{open}} = -\frac{1}{2} \frac{d^2}{d\omega^2} \text{Re } \tilde{K}(\omega) \Big|_{\omega=0}. \quad (4)$$

When the temporal moments

$$K_n = \int_0^\infty s^n K(s) ds$$

exist, expansion of the retarded response around $\omega = 0$ gives

$$m_{\text{open}} = \frac{K_2}{2} = \frac{1}{2} \int_0^\infty s^2 K(s) ds.$$

The controlled time-domain reduction and its remainder bound are given in Appendix A.

When the kernel admits a characteristic memory scale, let τ_K denote its effective support or decay time, and let ω_q characterize the retained motion. In the local regime

$$\omega_q \tau_K \ll 1,$$

substitution into Eq. (3) gives

$$\begin{aligned} m_{\text{eff}} \ddot{q} + \Gamma_{\text{eff}} \dot{q} + \kappa_{\text{eff}} q &= F_{\text{drive}}, \\ m_{\text{eff}} &= m_0 + m_{\text{open}}, \\ \Gamma_{\text{eff}} &= \Gamma_0 - K_1, \quad \kappa_{\text{eff}} = \kappa_0 + K_0. \end{aligned} \quad (5)$$

Thus the zeroth, first, and second temporal orders of the same causal response generate, respectively, static, first-order, and inertial sectors. For a translational retained coordinate, grouping the static and first-order terms into the resolved force gives the Newtonian form $F_{\text{net}} = m_{\text{eff}} \ddot{q}$.

The inverse susceptibility correspondingly takes the low-frequency form

$$\begin{aligned} \chi_q^{-1}(\omega) &= \kappa_{\text{eff}} - i\omega\Gamma_{\text{eff}} - m_{\text{eff}}\omega^2 + O(\omega^3), \\ m_{\text{eff}} &= -\frac{1}{2} \partial_\omega^2 \text{Re } \chi_q^{-1}(\omega) \Big|_{\omega=0}. \end{aligned} \quad (6)$$

Stiffness, damping, and inertia must therefore be extracted jointly from the complex response rather than inferred from a resonance shift alone.

Causality does not by itself guarantee differentiability at zero frequency, finite temporal moments, scale separation, or nonnegativity of the inertial sector. Long-tailed kernels may retain genuinely nonlocal dynamics [15], while oscillatory or active effective responses may generate negative or non-symmetric mass sectors [16, 17]. The matrix extension and the nonnegativity condition for a passive reciprocal inertial sector are summarized in Appendix A.

4. Passive spectral realization

A minimal harmonic exterior provides a passive realization in which the induced inertial sector is nonnegative. The model is used only as a spectral consistency test; it does not identify the cosmological exterior with an oscillator bath.

Consider a retained coordinate linearly coupled to external modes,

$$L = \frac{m_0}{2} \dot{q}^2 - \frac{\kappa_0}{2} q^2 + \sum_\alpha \left[\frac{m_\alpha}{2} \dot{x}_\alpha^2 - \frac{m_\alpha \Omega_\alpha^2}{2} x_\alpha^2 + g_\alpha q x_\alpha \right], \quad (7)$$

with $m_\alpha > 0$ and $\Omega_\alpha^2 > 0$. Retarded elimination of the external modes is summarized in Appendix B. Below the spectral gap,

$$|\omega| \ll \Omega_{\text{gap}}, \quad \Omega_{\text{gap}} \equiv \min_\alpha \Omega_\alpha,$$

the reduced inverse susceptibility is

$$\begin{aligned} \chi_q^{-1}(\omega) &= \kappa_{\text{ren}} - \omega^2 (m_0 + m_{\text{open}}) + O(\omega^4), \\ \kappa_{\text{ren}} &= \kappa_0 - \sum_\alpha \frac{g_\alpha^2}{m_\alpha \Omega_\alpha^2}, \\ m_{\text{open}} &= \sum_\alpha \frac{g_\alpha^2}{m_\alpha \Omega_\alpha^4} \geq 0. \end{aligned} \quad (8)$$

Nonnegativity of m_{open} follows from the positive-energy external modes and is strict whenever at least one mode is coupled. By contrast, stability of the coupled reference configuration requires the independent Schur-complement condition $\kappa_{\text{ren}} > 0$. The spectral definition is therefore applicable without requiring an absolutely convergent temporal moment for the undamped discrete environment.

For couplings $g_\alpha = \lambda \bar{g}_\alpha$, the bilinear model gives the exact scaling

$$m_{\text{open}}(\lambda) = \lambda^2 m_{\text{open}}^{(1)}.$$

If a single external mode dominates, then, at fixed modal mass and coupling strength,

$$m_{\text{open}} \propto \Omega^{-4},$$

whereas the associated static correction scales as Ω^{-2} . Stiffness and inertia must therefore be distinguished through the low-frequency curvature of the susceptibility rather than inferred from a resonance shift alone.

5. Generalized Mach principle and falsifiability

Generalized Mach principle. The local inertial response of a retained subsystem may contain a sector generated by its causal variational coupling to external degrees of freedom. Its strong form asserts that this induced sector exhausts the observed inertia.

The reduced theory accordingly separates the observed inertial coefficient as

$$m_{\text{obs}} = m_0 + m_{\text{open}}, \quad (9)$$

where m_0 is the residual local sector in the chosen reduced description. The weak Machian statement is the result established above: causal variational coupling to the exterior

can contribute to local inertial response. The strong Machian hypothesis adds

$$m_0 = 0.$$

This stronger claim does not follow from the memory reduction or from the passive spectral realization.

For the bilinear model, continuous reduction of the coupling gives

$$m_{\text{obs}}(\lambda) = m_0 + \lambda^2 m_{\text{open}}^{(1)}, \quad \lim_{\lambda \rightarrow 0} m_{\text{obs}}(\lambda) = m_0. \quad (10)$$

More general reductions invariant under $\lambda \rightarrow -\lambda$ may contain higher even-order corrections. A nonzero intercept falsifies the strong hypothesis within the realization and reduction being tested, while leaving the induced sector intact. A zero intercept is compatible with the strong hypothesis, but does not by itself identify the cosmological Universe as the physical source of the coupling.

Operationally, $m_{\text{obs}}(\lambda)$ is extracted through Eq. (6), with the retained coordinate, calibration, preparation protocol, and coarse-graining rule held fixed. In an effective theory, the residual-induced split may depend on this prescription. A universal strong Mach principle would instead require a fundamental derivation in which the zero-coupling intercept vanishes independently of the effective representation.

6. Minimal realizations and observables

Synthetic delay systems provide controllable realizations of the local memory-to-inertia mechanism. They test the reduced dynamical construction, not a cosmological origin of elementary-particle mass. For delayed mechanical feedback,

$$\begin{aligned} m_0 \ddot{q} + \Gamma_0 \dot{q} + \kappa_0 q + kq(t - \tau) &= F_{\text{drive}}, \\ m_{\text{open}} &= \frac{k\tau^2}{2}, \\ \Gamma_{\text{eff}} &= \Gamma_0 - k\tau, \\ \kappa_{\text{eff}} &= \kappa_0 + k. \end{aligned} \quad (11)$$

These local coefficients follow from

$$q(t - \tau) = q(t) - \tau \dot{q}(t) + \frac{\tau^2}{2} \ddot{q}(t) + O(\tau^3)$$

and are valid within the low-frequency regime

$$|\omega|\tau \ll 1.$$

For positive k , the first-order sector reduces the effective damping. Consequently, nonnegativity of the induced mass does not by itself ensure stability: the admissible operating regime must be determined from the spectrum of the complete delayed equation. Its characteristic equation is given in Appendix B. Delayed feedback is known to generate Hopf bifurcations and multistability [18]. Coherent-feedback experiments demonstrate controllable modifications of oscillator frequency and damping [19], while delay-based oscillator reductions provide explicit realizations of tunable effective inertia [20].

A delayed capacitive response gives the corresponding electrical analog. Let Q be the charge and $I = \dot{Q}$ the current. Then

$$\begin{aligned} V_{\text{fb}}(t) &= C^{-1} Q(t - \tau) \\ &= C^{-1} Q - \frac{\tau}{C} \dot{Q} + \frac{\tau^2}{2C} \ddot{Q} + O(\tau^3), \\ L_{\text{open}} &= \frac{\tau^2}{2C}. \end{aligned}$$

Because $\ddot{Q} = \dot{I}$, the second-order term has the form of an inductive voltage, $L_{\text{open}} \dot{I}$. Its sign, physical polarity, and stability remain properties of the complete feedback circuit rather than of the truncated coefficient alone.

Relative to the uncoupled coefficients $(\kappa_0, \Gamma_0, m_0)$, a mechanical realization at fixed feedback stiffness k predicts the hierarchy

$$\Delta\kappa = k, \quad \Delta\Gamma = -k\tau, \quad \Delta m = \frac{k\tau^2}{2}. \quad (12)$$

A joint fit of the real and imaginary parts of the inverse susceptibility should therefore recover delay-independent, linear, and quadratic contributions over the same low-frequency interval. This correlated temporal-order hierarchy is more restrictive than the observation of a resonance shift or an apparent mass change alone.

Dynamic and odd-mass metamaterials show more generally that force–acceleration coefficients can be emergent, frequency-dependent, and controllable [16, 17]. The specific prediction of the present framework is that a causal *coordinate*-memory response generates linked static, first-order, and inertial sectors. These realizations establish the structural portability and experimental accessibility of the mechanism; they do not by themselves establish its cosmological origin.

7. Discussion and conclusion

The central result of this Letter is conditional and precise. A causal coordinate response whose real frequency-domain kernel is twice differentiable at zero frequency defines an induced inertial coefficient through its low-frequency curvature. When the temporal second moment exists, the same coefficient is $m_{\text{open}} = K_2/2$. Variational openness supplies the structural origin of this response: it is the reduced contribution of exterior degrees of freedom whose variations cannot be discarded from the balance selecting the retained trajectory. The mechanism can therefore be summarized as

$$\begin{aligned} \text{open variational balance} &\implies \text{causal exterior response} \\ &\implies \text{low-frequency inertial sector.} \end{aligned}$$

This construction connects, but does not identify, three neighboring traditions. Projection and influence-functional methods derive memory after eliminating unresolved degrees of freedom [8, 9, 11, 12]. Machian programmes seek a relational origin of inertia through gravitational induction, scalar–tensor gravity, global absorber conditions, or relational configuration geometry [3–6]. Engineered feedback systems

and dynamic metamaterials produce controllable effective masses without necessarily assigning them a variational or Machian interpretation [16, 17, 19, 20]. The present contribution is the specific synthesis of these elements: coordinate-memory curvature is identified as induced inertia, traced to an open variational balance, separated from a residual local sector, and subjected to an operational decoupling test.

The framework does not specify the fundamental carrier of the exterior response. The interface $\partial\mathcal{S}$ denotes the variational partition between retained and complementary degrees of freedom, while $\mathcal{B}_{\mathcal{V} \rightarrow \mathcal{S}}$ denotes their contribution to the variational balance. Neither requires a material transmission medium or a local contact interaction. A covariant extension must derive the retarded kernel from explicit exterior fields, determine the conditions under which its curvature defines a positive inertial metric, and establish whether the zero-coupling intercept is independent of the retained variables and coarse-graining prescription. Earlier vacuum-based proposals illustrate both the attraction and the risk of assigning a specific microscopic carrier before these structural conditions have been established [21].

The limitations follow directly from the reduction. Causality alone does not guarantee differentiability at zero frequency, finite temporal moments, scale separation, nonnegativity, or dynamical stability. Long-tailed kernels may remain genuinely nonlocal; oscillatory or active responses may generate zero, negative, or nonsymmetric inertial sectors; and large delays invalidate the local derivative expansion. Laboratory realizations can test the memory-to-inertia mechanism and its zero-coupling intercept, but not by themselves establish a cosmological origin of elementary-particle mass.

Variational openness therefore does not prove that all inertia is Machian. It establishes a precise mechanism by which causal variational coupling to exterior degrees of freedom can generate an induced local inertial sector, provides a nonnegative passive realization, and separates this weak result from the strong hypothesis $m_0 = 0$. In this qualified sense, the observed inertia of a retained subsystem may contain the local low-frequency trace of its causal and variational openness to the larger physical system.

A. Controlled local reduction

Let $K : [0, \infty) \rightarrow \mathbb{R}$ be measurable and satisfy

$$\int_0^\infty (1 + s^3) |K(s)| ds < \infty.$$

This condition ensures the existence of the temporal moments K_n for $0 \leq n \leq 3$. Let q be three times continuously differentiable on the causal interval sampled by the kernel, with bounded third derivative there. Taylor's theorem gives, for each $s \geq 0$,

$$q(t-s) = q(t) - s\dot{q}(t) + \frac{s^2}{2}\ddot{q}(t) - \frac{s^3}{6}q^{(3)}(t - \theta_s s), \quad 0 < \theta_s < 1. \quad (13)$$

Substitution into the coordinate-memory functional yields

$$\mathcal{F}_K[q](t) = K_0 q(t) - K_1 \dot{q}(t) + \frac{K_2}{2} \ddot{q}(t) + R_3(t), \quad (14)$$

where

$$R_3(t) = -\frac{1}{6} \int_0^\infty s^3 K(s) q^{(3)}(t - \theta_s s) ds.$$

Defining the sampled causal interval

$$I_t = \{t - \theta s : s \in \text{supp } K, 0 \leq \theta \leq 1\},$$

the remainder satisfies

$$|R_3(t)| \leq \frac{1}{6} \sup_{u \in I_t} |q^{(3)}(u)| \int_0^\infty s^3 |K(s)| ds. \quad (15)$$

This bound guarantees finiteness of the remainder. Its smallness requires, in addition, separation between the characteristic memory time and the timescale of the retained motion.

The same integrability assumptions permit differentiation of the frequency response under the integral sign. With

$$\tilde{K}(\omega) = \int_0^\infty K(s) e^{i\omega s} ds,$$

one obtains

$$\frac{d^2}{d\omega^2} \text{Re } \tilde{K}(\omega) \Big|_{\omega=0} = -K_2,$$

which establishes the equivalence between the spectral and time-domain definitions of m_{open} whenever the second moment exists.

For vector-valued retained coordinates, define the symmetric part of the matrix kernel by

$$\mathbf{K}_{\text{sym}}(s) = \frac{1}{2} [\mathbf{K}(s) + \mathbf{K}^T(s)].$$

The reciprocal inertial sector is then

$$\mathbf{M}_{\text{open}} = -\frac{1}{2} \partial_\omega^2 \text{Re } \tilde{\mathbf{K}}_{\text{sym}}(\omega) \Big|_{\omega=0}. \quad (16)$$

When the matrix second moment exists,

$$\mathbf{M}_{\text{open}} = \frac{1}{2} \int_0^\infty s^2 \mathbf{K}_{\text{sym}}(s) ds.$$

A passive reciprocal inertial sector requires

$$\mathbf{v}^T \mathbf{M}_{\text{open}} \mathbf{v} \geq 0 \quad \text{for every real vector } \mathbf{v}.$$

Thus $\mathbf{M}_{\text{open}}$ is positive semidefinite. A positive-definite inertial metric would require strict inequality for every nonzero $\mathbf{v}$. This condition is additional to causality; the antisymmetric, nonreciprocal sector lies outside it.

B. Retarded elimination and delay stability

For the harmonic realization of Eq. (7), the retarded solution of each external mode is

$$x_\alpha(\omega) = \frac{g_\alpha q(\omega)}{m_\alpha [\Omega_\alpha^2 - (\omega + i0^+)^2]} + x_\alpha^{\text{hom}}(\omega). \quad (17)$$

The homogeneous contribution is fixed by the preparation of the external modes and is absorbed into the resolved drive. Substitution into the retained-coordinate equation gives

$$\chi_q^{-1}(\omega) = \kappa_0 - m_0 \omega^2 - \sum_\alpha \frac{g_\alpha^2}{m_\alpha [\Omega_\alpha^2 - (\omega + i0^+)^2]}. \quad (18)$$

For $|\omega| \ll \Omega_{\text{gap}}$, expansion of the external-mode denominators gives Eq. (8). The retarded prescription is retained in Eq. (18) because the exact response becomes resonant when the driving frequency approaches the external spectrum.

For the delayed mechanical realization, the exact inverse susceptibility is

$$\chi_{\text{delay}}^{-1}(\omega) = \kappa_0 - i\omega\Gamma_0 - m_0\omega^2 + ke^{i\omega\tau}.$$

For exponential modes $q(t) \propto e^{zt}$, the corresponding characteristic equation is

$$m_0 z^2 + \Gamma_0 z + \kappa_0 + ke^{-z\tau} = 0. \quad (19)$$

Exact asymptotic stability requires every characteristic root to satisfy

$$\text{Re } z < 0.$$

The second-order local truncation instead gives the effective coefficients

$$m_{\text{eff}} = m_0 + \frac{k\tau^2}{2}, \quad \Gamma_{\text{eff}} = \Gamma_0 - k\tau, \quad \kappa_{\text{eff}} = \kappa_0 + k.$$

For $m_{\text{eff}} \neq 0$, the Routh–Hurwitz conditions for the second-order truncation are

$$\frac{\Gamma_{\text{eff}}}{m_{\text{eff}}} > 0, \quad \frac{\kappa_{\text{eff}}}{m_{\text{eff}}} > 0.$$

On the ordinary positive-mass branch these reduce to

$$m_0 + \frac{k\tau^2}{2} > 0, \quad \Gamma_0 - k\tau > 0, \quad \kappa_0 + k > 0.$$

These are the stability conditions of the local second-order approximation; they do not replace the complete delay-spectrum test.

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Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No datasets were generated or analysed during the current study.

CRediT authorship contribution statement

Francisco Monroy: Conceptualization, Formal analysis, Methodology, Writing – original draft, Writing – review and editing.

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